Lecture 2. Linear Systems: LU Decomposition

Flops, triangular systems, Gaussian elimination, LU, pivoting, conditioning and stability

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2.1 Algorithms and their cost

We study algorithms for $Ax = b$. A guiding principle (Gentle):

Mantra

The form of a mathematical expression and the way the expression should be evaluated in practice may be quite different.

An **algorithm** is a finite set of instructions mapping input to output. Its cost is counted in **flops** (one floating-point $+, -, \times, \div$ or comparison, with fetch and store); it has order $O(f(n))$ if its leading flop count is $c\, f(n)$. E.g. Ab with $A \in \mathbb{R}^{m\times n}$ costs $2mn$ flops and AB with $B \in \mathbb{R}^{n\times p}$ costs $2mnp$.

At 10^{12} flops per second, $n = 10^{12}$ takes about 27 seconds with an $O(n \log n)$ algorithm and 31700 years with an $O(n^2)$ one; quicksort and Tukey's FFT turn $O(n^2)$ into $O(n \log n)$.

2.2 Triangular systems

General systems are reduced to easy ones. If A is lower triangular, **forward substitution** solves $Ax = b$:

$$x_1 = \frac{b_1}{a_{11}}, \qquad x_i = \frac{1}{a_{ii}}\Big(b_i - \sum_{j<i} a_{ij}x_j\Big), \quad i = 2, \dots, n,$$

and for upper triangular A, **back substitution** solves from x_n upwards. Each costs $1 + 3 + \cdots + (2n-1) = n^2$ flops, in place (b overwritten by x); BLAS routine `trsm`.

Triangular matrices have their diagonal entries as eigenvalues and their product as determinant; products and inverses of (unit) lower triangular matrices are (unit) lower triangular, and likewise upper. Products: for A, B upper triangular, $(AB)_{ij} = \sum_k a_{ik}b_{kj}$ needs $i \le k \le j$, so it vanishes for $i > j$ and equals $a_{ii}b_{ii}$ for $i = j$. Inverses: induct on n with $A = \left(\begin{smallmatrix} A_{11} & a_{12} \\ 0 & a_{22} \end{smallmatrix}\right)$; the block B_{21} of $B = A^{-1}$ satisfies $a_{22}B_{21} = 0$, so $B_{21} = 0$, $A_{11}B_{11} = I$, and $B_{11} = A_{11}^{-1}$ is upper triangular by induction (unit diagonal if A has one). Lower triangular cases follow by transposition. Hence an orthogonal triangular Q is diagonal, since Q^{-1} is upper and $Q^\top$ lower triangular.

2.3 Gaussian elimination and the LU decomposition

The **elementary operator matrix** $E_{jk}(c) = I + c\, e_j e_k^\top$ $(j \ne k)$ is unit triangular, $E_{jk}(c)^{-1} = E_{jk}(-c)$, and left-multiplication by it adds $c\times$row k to row j ($2m$ flops for m columns). Gaussian elimination reduces A to upper triangular form by such operations.

Those on column k combine into the **Gauss transformation**

$$M_k = E_{nk}(c_{nk}) \cdots E_{k+1,k}(c_{k+1,k}) = I + \mathbf{c}_k e_k^\top, \qquad \mathbf{c}_k = (0, \dots, 0, c_{k+1,k}, \dots, c_{nk})^\top,$$

with $c_{ik} = -a_{ik}^{(k)}/a_{kk}^{(k)}$, which zeros column k below the diagonal; it is unit lower triangular with $M_k^{-1} = I - \mathbf{c}_k e_k^\top$. Elimination computes

$$M_{n-1} \cdots M_1 A = U \quad \text{(upper triangular)}, \qquad \text{so} \qquad A = LU, \quad L = M_1^{-1} \cdots M_{n-1}^{-1}.$$

Proposition 2.1

$L = I - \sum_{k=1}^{n-1} \mathbf{c}_k e_k^\top$: it is unit lower triangular, and its (i,k) entry below the diagonal is the multiplier $\ell_{ik} = a_{ik}^{(k)}/a_{kk}^{(k)}$.

Proof. $M_k^{-1}M_{k+1}^{-1} = (I - \mathbf{c}_k e_k^\top)(I - \mathbf{c}_{k+1}e_{k+1}^\top) = I - \mathbf{c}_k e_k^\top - \mathbf{c}_{k+1}e_{k+1}^\top + \mathbf{c}_k(e_k^\top \mathbf{c}_{k+1})e_{k+1}^\top$, and $e_k^\top \mathbf{c}_{k+1} = 0$ because $\mathbf{c}_{k+1}$ vanishes in positions $\le k+1$. Inductively the product is $I - \sum_k \mathbf{c}_k e_k^\top$. ■

Theorem 2.2 (Existence and uniqueness of LU)

If the leading principal submatrices $A[1{:}k, 1{:}k]$, $k = 1, \dots, n-1$, are nonsingular, then $A = LU$ with L unit lower triangular and U upper triangular. If moreover A is nonsingular, the decomposition is unique and $\det A = \prod_k u_{kk}$.

Proof. *Existence.* Unit lower triangular operations preserve leading principal minors, so the step-k pivot is $a_{kk}^{(k)} = \det A[1{:}k, 1{:}k] / \det A[1{:}k-1, 1{:}k-1] \neq 0$ and elimination completes. *Uniqueness.* If $L_1U_1 = L_2U_2$ with A nonsingular, all factors are invertible and $L_2^{-1}L_1 = U_2U_1^{-1}$ is both unit lower and upper triangular, hence I. Finally $\det A = \det L \det U = \prod u_{kk}$ (how software computes determinants). ■

For nonsingular A the condition is also necessary: A has an LU decomposition without pivoting *iff* all leading principal submatrices are nonsingular. Indeed, partitioning conformally, $A[1{:}k, 1{:}k] = L_kU_k$ with L_k, U_k the leading blocks, and $0 \neq \det A = \prod_i u_{ii}$ gives $\det A[1{:}k, 1{:}k] = \prod_{i \le k} u_{ii} \neq 0$.

Cost. Step k updates an $(n-k) \times (n-k)$ block at 2 flops per entry, so LU costs

$$\sum_{k=1}^{n-1} 2(n-k)^2 \approx \tfrac{2}{3}n^3 \text{ flops},$$

in place (L below the diagonal, U on and above). Solving $LUx = b$ adds $2n^2$ (forward and back substitution), $\frac{2}{3}n^3 + 2n^2$ in total; several right-hand sides reuse the factorization.

No inversion. Inverting A via LU solves n systems with right-hand sides $e_1, \dots, e_n$, costing $\frac{4}{3}n^3$ (exploiting zeros), $2n^3$ in total: three times one solve. Rule: *for $A^{-1}b$, solve $Ax = b$.* Explicit inverses are needed only for standard errors, very many right-hand sides, or small n.

2.4 Pivoting, conditioning and stability

For $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ elimination fails at once (zero pivot); small pivots are just as harmful.

Example 2.1

Solve $0.001x_1 + x_2 = 1$, $x_1 + x_2 = 2$ (exact solution $\approx (1.001, 0.999)$) with two significant digits. Pivot 0.001 gives multiplier 1000 and $(1 - 1000)x_2 = 2 - 1000$, rounded to $-1000x_2 = -1000$: $x_2 = 1$, $x_1 = 0$. The computed $L = \begin{pmatrix} 1 & 0 \\ 1000 & 1 \end{pmatrix}$, $U = \begin{pmatrix} 0.001 & 1 \\ 0 & -1000 \end{pmatrix}$ give $LU = \begin{pmatrix} 0.001 & 1 \\ 1 & 0 \end{pmatrix} \neq A$: the large multiplier wiped out a_{22} and b_2. Swapping the equations first gives multiplier 0.001, $(1 - 0.001)x_2 = 1 - 0.002$, so $x_2 = 1$, $x_1 = 1$, correct to two digits.

Conditioning. The *problem* $A \mapsto A^{-1}b$ is **well conditioned** if small relative perturbations of A cause small relative changes of the solution, as measured by the **condition number**

$$\kappa(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}, \qquad \sigma_{\max}(A) = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}, \quad \sigma_{\min}(A) = \min_{x \neq 0} \frac{\|Ax\|}{\|x\|},$$

(∞ for singular A). In Example 2.1, $\kappa(A) = 2.62$: well conditioned.

Example 2.2 (An ill-conditioned matrix)

The symmetric matrix

$$A = \begin{pmatrix} 5 & 7 & 6 & 5 \\ 7 & 10 & 8 & 7 \\ 6 & 8 & 10 & 9 \\ 5 & 7 & 9 & 10 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} 68 & -41 & -17 & 10 \\ -41 & 25 & 10 & -6 \\ -17 & 10 & 5 & -3 \\ 10 & -6 & -3 & 2 \end{pmatrix},$$

looks harmless. Changing a_{33} from 10 to 9.99 (relative change 10^{-3}) gives the inverse (two decimals) $\begin{pmatrix} 71.04 & -42.79 & -17.89 & 10.54 \\ -42.79 & 26.05 & 10.53 & -6.32 \\ -17.89 & 10.53 & 5.26 & -3.16 \\ 10.54 & -6.32 & -3.16 & 2.09 \end{pmatrix}$, entries moving by up to 5.4%: an amplification of about 54. The worst direction is much worse: $\kappa_2(A) = \sigma_{\max}/\sigma_{\min} \approx 2984$, so about 3.5 decimal digits can be lost in solving $Ax = b$.

Stability. An *algorithm* $\tilde{f}$ is **stable** if for every $\tilde{A}$ within relative distance ϵ of A, the computed $\hat{x} = \tilde{f}(A)$ is within relative distance $O(\epsilon)$ of $\tilde{A}^{-1}b$. By Example 2.1, elimination without pivoting is unstable: a well-conditioned problem gives $\|x - \hat{x}\|/\|x\| \geq 1/\sqrt{2}$.

Partial pivoting. Before eliminating column k, swap into row k the row with the largest $|a_{ik}|$, $i \geq k$; at $O(n^2)$ comparisons this gives

$$PA = LU, \qquad P \text{ a permutation matrix}, \quad |\ell_{ij}| \leq 1.$$

Complete pivoting also swaps columns, moving the largest remaining entry to the pivot: $PAQ = LU$, more stable but costlier. Gaussian elimination with partial pivoting (GEPP) is the standard dense solver and is stable in practice, partly because $|\ell_{ij}| \leq 1$.

Software. BLAS (basic operations) and LAPACK (decompositions): `getrf` computes $PA = LU$ in place; R's `solve` calls `dgesv`.

2.5 Selected exercises

Motivation. One LDU factorization by hand, then three facts used without proof: the condition-number error bound, that partial pivoting never breaks down, and when no pivoting is needed.

Exercise 2.1 (An LDU decomposition by hand)

Let

$$A = \begin{pmatrix} 2 & 4 & 2 \\ 4 & 7 & 7 \\ -2 & -7 & 5 \end{pmatrix}, \qquad b = \begin{pmatrix} 4 \\ 13 \\ 7 \end{pmatrix}.$$

(a) Compute $A = LU$ by Gaussian elimination, and write it as $A = LDU_1$ with D diagonal and U_1 unit upper triangular.
(b) Solve $Ax = b$ by forward and back substitution, and compute $\det A$.

Solution.

(a) The leading principal minors are 2, $2 \cdot 7 - 4 \cdot 4 = -2$ and 4 (below), so by Theorem 2.2 the factorization exists and is unique. Column 1 ($\ell_{21} = 2$, $\ell_{31} = -1$):

$$\begin{pmatrix} 2 & 4 & 2 \\ 0 & -1 & 3 \\ 0 & -3 & 7 \end{pmatrix}.$$

Column 2 ($\ell_{32} = 3$) leaves $(0, 0, -2)$ in row 3. By Proposition 2.1,

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 4 & 2 \\ 0 & -1 & 3 \\ 0 & 0 & -2 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \quad U_1 = D^{-1}U = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}.$$

Check: rows 2, 3 of LU are $2(2,4,2) + (0,-1,3) = (4,7,7)$ and $-(2,4,2) + 3(0,-1,3) + (0,0,-2) = (-2,-7,5)$.

(b) $Ly = b$: $y_1 = 4$, $y_2 = 13 - 2 \cdot 4 = 5$, $y_3 = 7 + 4 - 3 \cdot 5 = -4$. $Ux = y$: $x_3 = 2$, $x_2 = 3x_3 - 5 = 1$, $x_1 = (4 - 4x_2 - 2x_3)/2 = -2$. So $x = (-2, 1, 2)^\top$, and $Ax = (-4+4+4,\ -8+7+14,\ 4-7+10)^\top = b$. Finally $\det A = \prod_k u_{kk} = 2 \cdot (-1) \cdot (-2) = 4$. ■

Exercise 2.2 (The condition number bounds the error)

Let A be nonsingular, $b \neq 0$ and $Ax = b$. Write $\|\cdot\|$ for the Euclidean norm and the induced matrix norm, $\|M\| = \sigma_{\max}(M)$.
(a) Show that $\kappa(A) = \|A\|\,\|A^{-1}\|$.
(b) If $A(x + \delta x) = b + \delta b$, show that $\dfrac{\|\delta x\|}{\|x\|} \leq \kappa(A) \dfrac{\|\delta b\|}{\|b\|}$.
(c) If $(A + \delta A)(x + \delta x) = b$ and $t = \kappa(A)\|\delta A\|/\|A\| < 1$, show that $\dfrac{\|\delta x\|}{\|x\|} \leq \dfrac{t}{1-t}$.

Solution. [Supplement]

(a) $y = Ax$ ranges over nonzero vectors with x, so

$$\sigma_{\min}(A) = \min_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \min_{y \neq 0} \frac{\|y\|}{\|A^{-1}y\|} = \Big(\max_{y \neq 0} \frac{\|A^{-1}y\|}{\|y\|} \Big)^{-1} = \frac{1}{\|A^{-1}\|},$$

and $\kappa(A) = \sigma_{\max}(A)/\sigma_{\min}(A) = \|A\|\|A^{-1}\|$.

(b) $\delta x = A^{-1}\delta b$ gives $\|\delta x\| \leq \|A^{-1}\|\|\delta b\|$, and $\|b\| \leq \|A\|\|x\|$ ($x \neq 0$ as $b \neq 0$) gives $1/\|x\| \leq \|A\|/\|b\|$. Multiply and use (a).

(c) $A\delta x = -\delta A(x + \delta x)$, so $\delta x = -A^{-1}\delta A(x + \delta x)$ and

$$\|\delta x\| \leq \|A^{-1}\|\|\delta A\|(\|x\| + \|\delta x\|) = t(\|x\| + \|\delta x\|),$$

as $\|A^{-1}\|\|\delta A\| = t$ by (a). Rearranging, $(1 - t)\|\delta x\| \leq t\|x\|$ with $1 - t > 0$.

So relative data perturbations ϵ (rounding: $\epsilon \approx 10^{-16}$) give relative errors about $\kappa(A)\epsilon$: $\kappa(A) \approx 10^k$ loses about k digits, as in Example 2.2. ■

Exercise 2.3 (Partial pivoting never breaks down)

Show that for every nonsingular $A \in \mathbb{R}^{n \times n}$, Gaussian elimination with partial pivoting runs to completion and yields $PA = LU$ with P a permutation matrix, L unit lower triangular with $|\ell_{ij}| \leq 1$, and U upper triangular and nonsingular.

Solution. [Supplement]

Step 1. (The pivots are nonzero.) Suppose steps $1, \dots, k-1$ have been carried out, giving

$$M_{k-1}P_{k-1}\cdots M_1P_1A = A^{(k)} = \begin{pmatrix} U_{11} & U_{12} \\ 0 & S_k \end{pmatrix},$$

with P_j swapping rows j and some $j' \geq j$, $M_j = I + \mathbf{c}_j e_j^\top$, and U_{11} upper triangular of size $k-1$. All factors are nonsingular, so $0 \neq \det A^{(k)} = \det U_{11} \det S_k$; the first column of S_k is nonzero, its largest-modulus entry is a nonzero pivot, and each multiplier is an entry over one of larger or equal modulus: $|\ell_{ik}| \leq 1$. By induction elimination completes with $U = A^{(n)}$ upper triangular and nonsingular.

Step 2. (Moving permutations right.) Put $Q_k = P_{n-1}\cdots P_{k+1}$ ($Q_{n-1} = I$) and $N_k = Q_k M_k Q_k^{-1}$. Each P_j is a symmetric involution, so

$$M_{n-1}P_{n-1}M_{n-2}P_{n-2}\cdots M_1P_1 = N_{n-1}N_{n-2}\cdots N_1\,P, \qquad P = P_{n-1}\cdots P_1,$$

by inserting $P_jP_j = I$; e.g. for $n = 4$,

$$M_3P_3M_2P_2M_1P_1 = M_3\,(P_3M_2P_3)\,(P_3P_2M_1P_2P_3)\,(P_3P_2P_1).$$

Step 3. (Structure of N_k.) P_j, $j > k$, swaps rows $j, j' > k$, so $Q_k e_k = e_k$, $e_k^\top Q_k^{-1} = e_k^\top$ and $N_k = I + \tilde{\mathbf{c}}_k e_k^\top$ with $\tilde{\mathbf{c}}_k = Q_k\mathbf{c}_k$ a permutation of $\mathbf{c}_k$ within positions $k+1, \dots, n$: zero in positions $\leq k$, entries of modulus ≤ 1.

Step 4. (Conclusion.) $N_{n-1}\cdots N_1PA = U$, and the proof of Proposition 2.1 applies verbatim: $L = N_1^{-1}\cdots N_{n-1}^{-1} = I - \sum_k \tilde{\mathbf{c}}_k e_k^\top$ is unit lower triangular, $|\ell_{ij}| \leq 1$, and $PA = LU$. ■

Exercise 2.4 (Diagonal dominance makes pivoting unnecessary)

Suppose A is strictly diagonally dominant by columns: $|a_{jj}| > \sum_{i\neq j}|a_{ij}|$ for every j. Show that Gaussian elimination without pivoting runs to completion with all multipliers $|\ell_{ik}| < 1$, so that partial pivoting would never swap rows.

Solution. [Supplement]

Step 1. (One step.) Partition $A = \left(\begin{smallmatrix} \alpha & w^\top \\ v & B \end{smallmatrix}\right)$. Dominance of column 1 gives $\sum_i |v_i| < |\alpha|$, so $\alpha \neq 0$ and $|v_i/\alpha| < 1$. The remaining block is the Schur complement $S = B - vw^\top/\alpha$, $s_{ij} = b_{ij} - v_iw_j/\alpha$.

Step 2. (S inherits dominance.) For column j: $|b_{jj}| > \sum_{i\neq j}|b_{ij}| + |w_j|$, and Step 1 gives $\sum_{i\neq j}|v_i| < |\alpha| - |v_j|$. Hence

$$\begin{aligned}\sum_{i\neq j}|s_{ij}| &\leq \sum_{i\neq j}|b_{ij}| + \frac{|w_j|}{|\alpha|}\sum_{i\neq j}|v_i| < |b_{jj}| - |w_j| + \frac{|w_j|}{|\alpha|}(|\alpha| - |v_j|) \\ &= |b_{jj}| - \frac{|w_j||v_j|}{|\alpha|} \leq \left|b_{jj} - \frac{v_jw_j}{\alpha}\right| = |s_{jj}|.\end{aligned}$$

(Strictness comes from the bound on $|b_{jj}|$; the bound on $\sum|v_i|$ enters with the factor $|w_j| \geq 0$.)

Step 3. (Induction.) S is strictly column dominant of size $n-1$, so every pivot is nonzero and every multiplier has modulus < 1. Each pivot exceeds in modulus every other entry of its column below it, so partial pivoting always picks the diagonal entry. ■